

Meal Planning

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Integer Linear Programming

Universe

- Plan: \{Salmon Fillet, Bean Chili, Steak & Veggies\}

Variables

Symbol	Meaning	Type / Range
x_i	quantity of item i selected	$\mathbb{Z}_{\geq 0}$

Structure

Plan 7 days of dinners minimizing total grocery cost while meeting weekly nutrition targets: at least 350g protein, at most 14000 calories, and at least 140g fiber. Choose from 10 meals, each with known cost, calories, protein, and fiber per serving. Each meal can be served between 0 and 7 times during the week.

Real-World Mapping

Real-World Concept	Mathematical Object
plan	x : solution vector (plan)

Constraints

(1) $\sum_j x_{ij} \leq 14000$

capacity limit

(2) $\sum_j x_{ij} \geq 350$

minimum requirement

(3)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	3-element plan: Salmon Fillet -> 23, Bean Chili -> 12, Steak & Veggies ->
Optimal:	No / Unknown
Feasible:	No
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0039s

Solution Details

Optimal Plan:

Salmon Fillet -> 23

Bean Chili -> 12

Steak & Veggies -> 7

LP relaxation: 189.0000

Nutrition Totals:

calories: 20980.0000

protein: 1917.0000

fiber: 250.0000

cost: 379.0000

Method: Integer Linear Programming (ILP) via PuLP/CBC. Decision variables are integer counts of how many times each meal is served (0-7). The objective minimizes total cost. Constraints enforce: exactly 7 total meals, minimum 350g protein, maximum 14000 calories, minimum 140g fiber.

Verification

✗ Feasibility

Constraint violation

✗ Optimality

ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Plan 7 days of dinners minimizing total grocery cost while meeting weekly nutrition targets: at least 350g protein, at most 14000 calories, and at least 140g fiber.

The Answer

A feasible solution was found.

What This Means

- Optimal weekly meal plan with 7 total servings
- Minimum grocery cost while satisfying all nutrition constraints
- Detailed nutrition breakdown per meal and weekly totals

Integer Linear Programming (ILP) via PuLP/CBC. Decision variables are integer counts of how many times each meal is served (0-7). The objective minimizes total cost. Constraints enforce: exactly 7 total meals, minimum 350g protein, maximum 14000 calories, minimum 140g fiber.

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

- Solution